

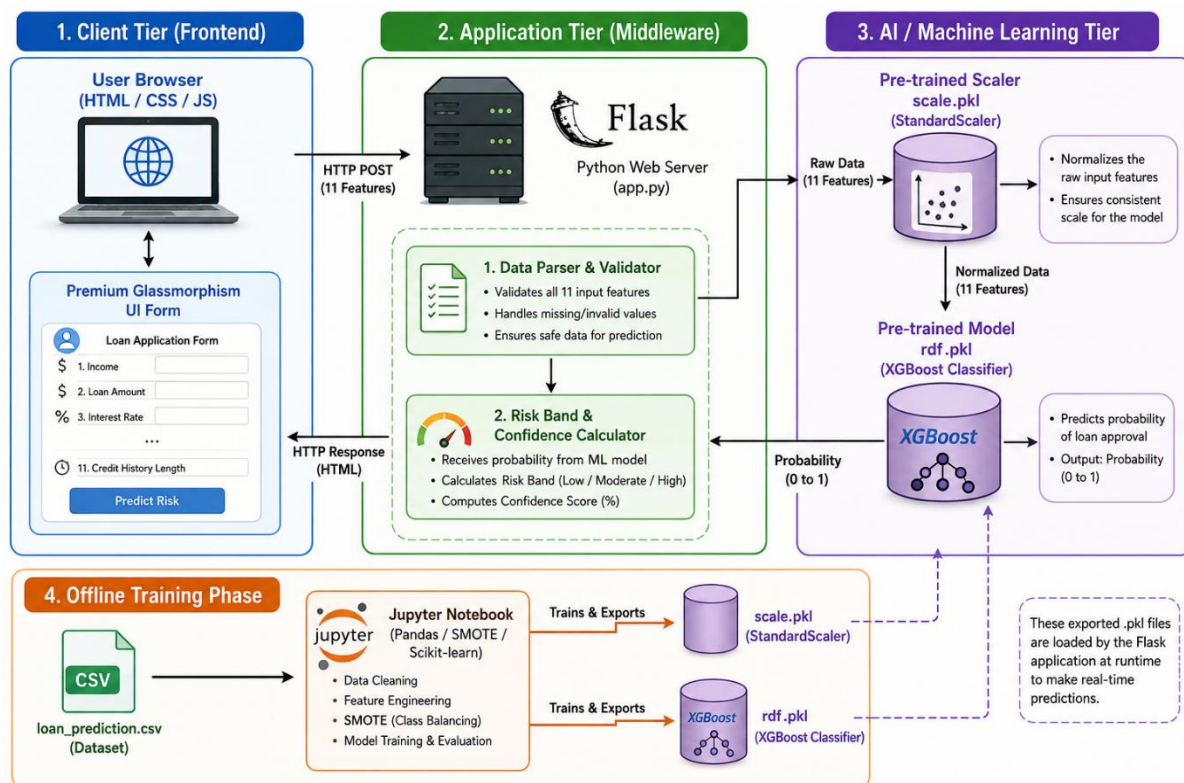
Solution Architecture

Date	27 June 2026
Team ID	
Project Name	Smart Lender
Maximum Marks	5 Marks

Solution Architecture Diagram:

The proposed solution architecture follows a modular four-tier design consisting of the Client Tier, Application Tier, AI/Machine Learning Tier, and Offline Training Tier. This architecture separates the user interface, business logic, prediction engine, and model training process, making the system scalable, maintainable, and efficient.

The frontend collects user inputs and sends them to the Flask backend, where the data is validated and processed before being passed to the pre-trained StandardScaler and XGBoost model for prediction. The prediction results, including approval probability and risk level, are then returned to the user, while the machine learning models are developed and updated separately using the training dataset in the offline training phase



Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer (Client Tier)	Provides the user interface for entering the 11 loan applicant features and displaying the prediction result, confidence score, and risk band.	HTML, CSS, JavaScript
Application Layer (Flask Backend)	Receives user input, validates and processes the data, communicates with the machine learning models, computes the risk band, and returns the prediction results to the frontend.	Python, Flask
AI / Machine Learning Layer	Preprocesses input using the trained StandardScaler and predicts loan approval probability using the pre-trained XGBoost model.	Scikit-learn (StandardScaler), XGBoost, Pickle (.pkl files)
Offline Training Layer	Performs data preprocessing, feature engineering, class balancing using SMOTE, model training, evaluation, and exports the trained models for deployment.	Jupyter Notebook, Pandas, NumPy, Scikit-learn, SMOTE, XGBoost